Anomalous paleomagnetism of the Crowsnest Formation of the Rocky Mountains[1]

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The volcanic Crowsnest Formation of Albian age (late Early Cretaceous) from the Rocky Mountain fold and thrust belt of Alberta has a stable remanent magnetization with a mean direction of 349°, 59° (α_{95} = 5°) and paleopole at 78°N, 108°E (dm = 7°, dp = 5°). The inclination is lower than, and the declination clockwise of, the expected mid-Cretaceous paleogeomagnetic field for cratonic North America. Taken at face value the result indicates that the Crowsnest Formation and the thrust sheet in which it occurs have been transported from the south relative to cratonic North America by 17 ± 6° (about 1800 km) and rotated 24 ± 10° clockwise. It is also possible that flattening of inclination is caused by magnetic anistropy, but tests show this to be unlikely. A third possibility is that the magnetization is secondary and of latest Cretaceous age, but there are good reasons for believing this is not so. Lastly, it is possible that the unit could have been formed close to its present position relative to the craton but was deposited so quickly that the paleosecular variation was not adequately sampled, and the result is only a "spot" reading of the paleofield. The last is our preferred interpretation of the flattened inclination, but the clockwise deflection of the declination could reflect rotation. Other paleomagnetic data from the fold and thrust belt are generally consistent with the third interpretation.

La formation de Crowsnest composée de roches volcaniques d'âge albien (Crétacé tardi-inférieur) de la zone de charriages des Rocheuses de l'Alberta possède une aimantation rémanente stable avec une direction moyenne de 349°, 59° (α_{95} = 5°) et un paléopole à 78°N, 108°E (dm = 7°, dp = 5°). L'inclinaison est plus faible que celle qu'on pourrait envisager pour un champ paléomagnétique du Crétacé moyen pour le craton de l'Amérique du Nord, et la déclinaison est tournée dans le sens des aiguilles d'une montre. En regard de ces résultats, la formation de Crowsnest et la nappe de charriage dans laquelle se trouve cette formation ont du être transportées du sud de 17 ± 6° (environ 1800 km) relativement au craton de l'Amérique du nord et tournées de 24 ± 10° dans le sens des aiguilles d'une montre. Il est possible que la diminution de l'inclinaison résulte de l'anisotropie magnétique, cependant des essais ont démontré que c'est peu probable. Une troisième hypothèse serait que l'aimantation soit secondaire et plus tardive que Crétacé, mais il y a d'excellentes raisons de croire que ce n'est pas le cas. Enfin, il est possible que l'unité aurait pu être formée très près de sa position actuelle par rapport au craton, mais qu'elle aurait été déposée si rapidement que la variation paléoséculaire n'ait pas été échantillonnée adéquatement et que le résultat ne serait qu'une lecture «ponctuelle» du paléochamp. Cette dernière interprétation pour expliquer la diminution de l'inclinaison est celle que nous préférons, cependant la déviation dans le sens des aiguilles d'une montre de la déclinaison peut témoigner de la rotation. D'autres données paléomagnétiques de la zone de charriages des Rocheuses s'accordent avec la troisième interprétation.

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Introduction

The Crowsnest Formation occurs in the foreland fold and thrust belt of southwestern Alberta (Figs. 1, 6). It is contained in rock panels that are detached from Precambrian basement and separated from one another by large, westerly-dipping thrusts. The formation consists of pyroclastic and epiclastic rocks with minor flows, all of trachytic composition (Norris 1964). It is overlain disconformably by the Upper Cretaceous Blackstone Formation. At its base the Crowsnest Formation interfingers with the underlying Lower Cretaceous Blairmore Group. It has a maximum thickness of 320 m and thins laterally. Flora of the upper part of the Blairmore Group is Albian in age, which can be taken to be the age of the Crowsnest Formation. Radiometric studies have provided imprecise indications of age. Whole-rock and mineral K/Ar ages (reviewed in Norris (1964)) vary from 61 to 126 Ma, a range beyond that permitted stratigraphically. Although discrete depositional units can be recognized, the total time represented is not known. It could be the product of only one eruption, and the paleomagnetic record could span only a few years. It could be the product of several eruptions spanning thousands of years, and the record could be sufficient to average out paleosecular variation, providing an estimate of the geocentric axial dipole (GAD) field direction. As we shall see, the uncertainty in the length of the paleomagnetic record is crucial to the discussion.

The generally accepted geological interpretation (see, for example, Norris (1964) and Price (1981)) is that the sedimentary rocks of the Rocky Mountains were laid down along the western margin of cratonic North America and then were thrust onto the craton as overlapping thrust sheets in the Laramide Orogeny sometime during the latest Cretaceous or Paleocene. According to this interpretation, the thrust sheets moved from the west or south-southwest by as much as 100 or 200 km but not by thousands of kilometres. Motions of hundreds of kilometres are not detectable paleomagnetically, so paleomagnetic inclinations should conform to those observed on the craton, but, since thrust sheets can rotate during emplacement, it is not unreasonable to expect that the paleomagnetic declination would be deflected, as Grubbs and van der Voo (1976) have observed in the fold and thrust belt of Idaho and Wyoming (see below).

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TABLE 1. Results from Iron Ridge section

Group	C, S	M	$D°$, $I°$	k	$\alpha_{95}°$	λ°N, φ°E	dp°, dm°
A(b)	*22, 35*	*0.41*	*349, 59*	*44*	*5*	*78, 108*	*5, 7*
A(h)	—	—	038, 40	—	—	50, 004	—
B	15, 17	0.51	331, 60	14	10	68, 114	11, 15
A + *B*	37, 52	0.45	342, 60	25	4	75, 136	5, 6
E	11, 22	0.43	332, 62	69	6	70, 148	7, 9
A + *B* + *E*	59, 74	0.44	339, 61	28	3	74, 134	4, 5
Expected	—	—	325, 71	—	—	68, 186	3, 3

NOTES: C is the number of cores, except in E row, where it refers to hand samples; S is the number of specimens; M is the mean magnetization ($\times 10^{-1}$ A m^{-1}); D,I are declination and inclination; k is Fisher's precision; and α_{95} is the radius of the circle of confidence ($P = 0.05$). Paleopoles (lat. °N, long. °E) and errors are shown on the right. Cores are given unit weight. A and B are the magnetization types as explained in the text; A(b) is with respect to bedding; A(h) is with respect to present horizontal. The E result is from samples collected (cleaned at 80 mT) in 1968. We consider A(b) to be the best estimate, and for this reason that row is set in italics. The last row gives the mean direction and paleopole calculated from North American craton data for the mid-Cretaceous (see legend to Fig. 5).

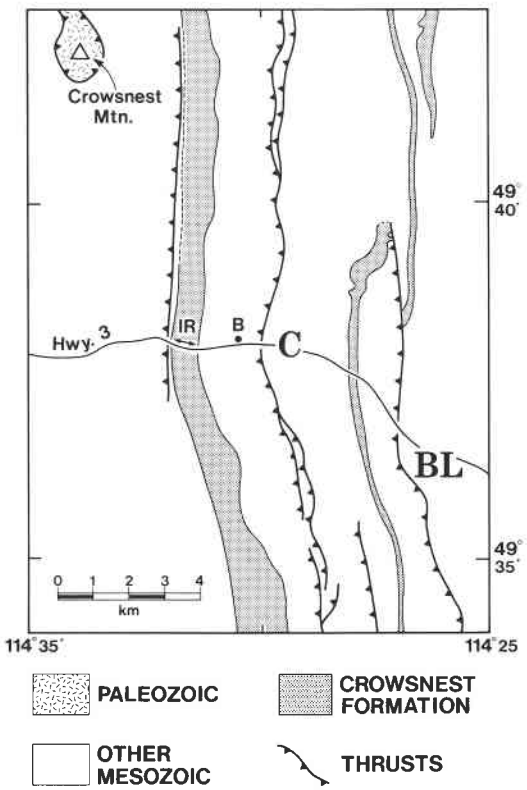


FIG. 1. Map of sampling area. Geology after Price (1962). IR: Iron Ridge section; B: location of Blairmore Group samples; C, BL: towns of Coleman and Blairmore. The Lewis Thrust is situated beneath Paleozoic rocks of Crowsnest Mountain.

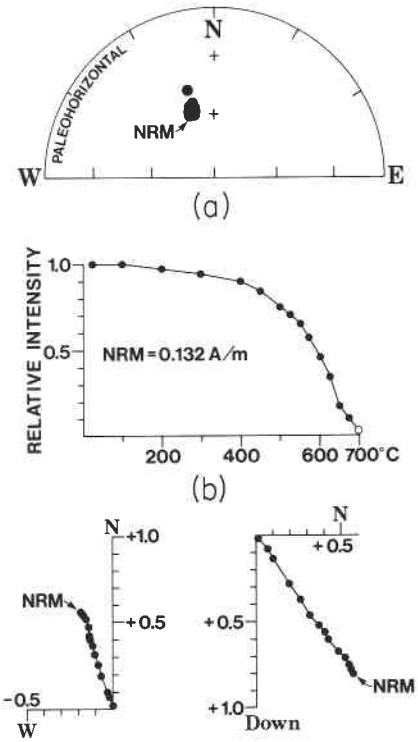


FIG. 2. Thermal demagnetization of A-type magnetization. (*a*) Changes in direction. (*b*) Changes in intensity. (*c*) Normalized orthogonal plots.

A collection of 11 hand samples was made in 1968. After magnetic cleaning in 80 mT it was found that the magnetizations were directed to the north, with a downward inclination of about 60° (Table 1, entry E). This value is lower than that expected (71°) for the mid-Cretaceous and indicates a displacement from the south in excess of 1000 km, much greater than that proposed by geological interpretations. In order to check this anomalous result, further collections were made in 1981 and 1982 to repeat the observations from the Crowsnest Formation itself and to extend the stratigraphic coverage to determine if the anomaly arose from failure to average out paleosecular variation.

New Results

Drill-core samples, oriented by magnetic and sun compasses, were collected from three localities, under the guidance of D. K. Norris of the Geological Survey of Canada: pyroclastic and flow rocks from Iron Ridge (48°38′N, 114°32′W), tuffs, probably waterlaid, at Star Creek (49°37′N, 114°33′W), and at Lynx Creek (49°28′N, 114°25′W). Other localities were visited, but all were deeply weathered and fractured. Coherent magnetizations were observed only from the Iron Ridge locality just west of Coleman, Alberta, where there is a superb new 0.8 km long road cutting, exposing the full thickness of the Crowsnest Formation (Fig. 1). The samples span about 300 m stratigraphically. All results reported here are from the Iron Ridge section. The

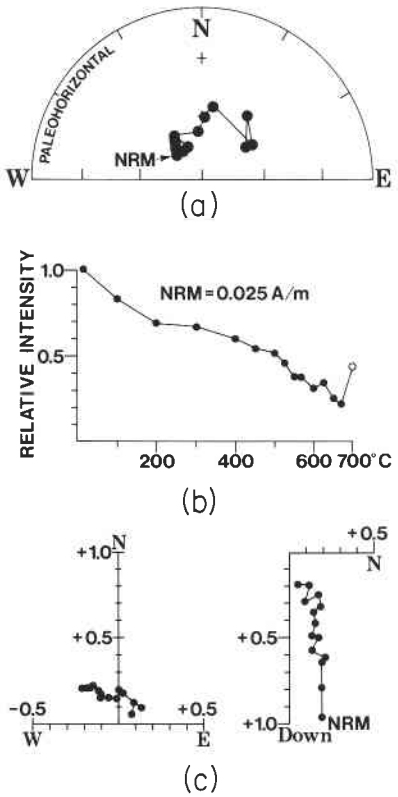


FIG. 3. Thermal demagnetization of *B*-type magnetization. (*a*) Changes in direction. (*b*) Changes in intensity. (*c*) Orthogonal plots.

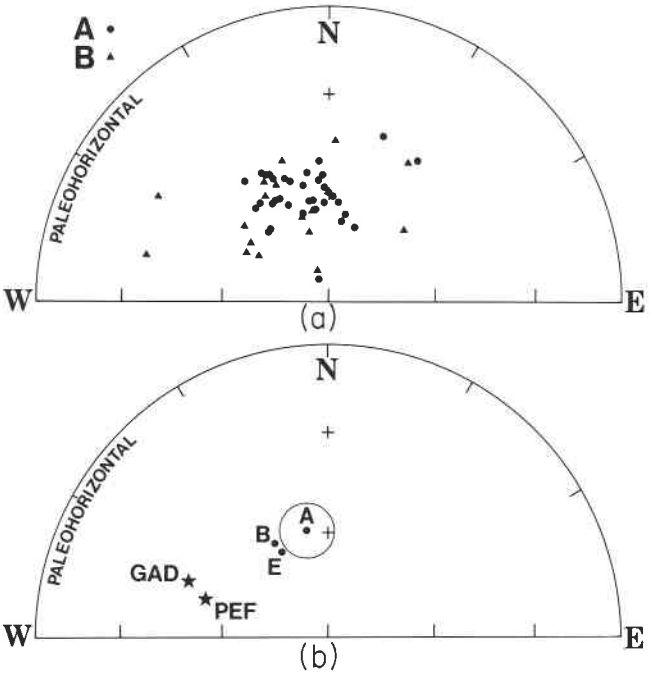


FIG. 4. Summary of the directions of magnetizations. (*a*) All specimen directions. (*b*) Mean directions, labelled as in Table 1. GAD: direction of the geocentric axial dipole field. PEF: direction of the present geomagnetic field. All directions are corrected to paleohorizontal using a geological dip of 42° at 260°.

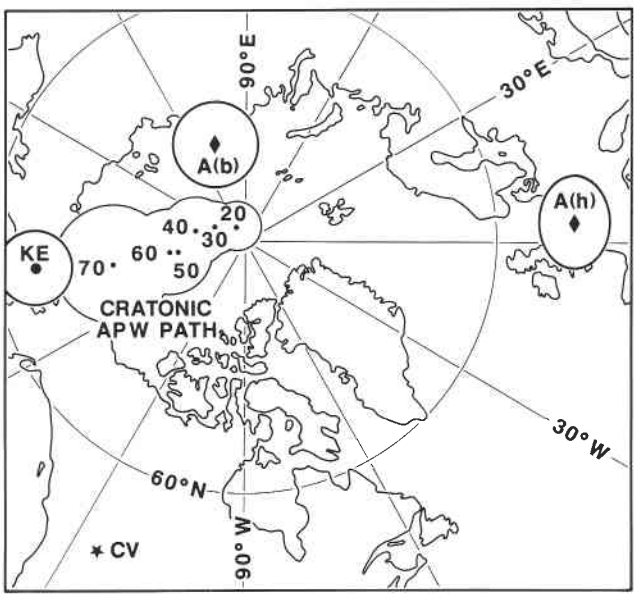


FIG. 5. The Crowsnest paleopole compared with that for the mid-Cretaceous of cratonic North America (A(b) corrected for tilt, A(h) with respect to present horizontal). KE: reference paleopole (68°N, 174°W, $\alpha_{95} = 3°$), an average of six cratonic studies listed in Irving and Irving (1982, Table 1, entries 39–44, age range 78–120 Ma, with a mean of 94 Ma). Reference paleopoles for 70–20 Ma are also from Irving and Irving (1982). CV: position of the Crowsnest Formation.

attitude (dip of 42° in a direction 260°E of N) is well defined from underlying sediments and local structure.

All specimens were subjected to detailed thermal or alternating field (AF) demagnetization. Two types of behaviour (*A* and *B*) were recognized (Figs. 2, 3). The *A*-type magnetization is univectorial, with no evidence of a second magnetization. The directions point slightly west of north and have a downward inclination of about 60°, like those obtained from the 1968 collections. Unblocking temperatures range from 400 to 700°C, and the orthogonal plots decay linearly to the origin (Fig. 2). Directions were estimated by choosing a value on the linear portion of the decay plot (about 60 mT for AF treatment, or between 300 and 525°C for thermal treatment). The directions are well grouped (Fig. 4*a*). The *A* magnetization is typically found in samples of grey, fresh lahar or flow rock. Study of polished sections revealed abundant, fine-grained opaque minerals 1 μm or so in diameter, some of which occur as alteration products of silicate minerals. Larger particles are present as euhedral grains of magnetite ($\sim$20 μm in diameter), irregular grains of magnetite and hematite, and composite grains of magnetite, hematite, and ilmenite.

In the second type (*B*-type) of behaviour the orthogonal plots are less regular and do not decay to the origin. Initially the directions point to the north and down, but commonly they migrate towards the east at high temperature without reaching an end point. This shows that a magnetization remains at high temperatures, but it cannot be isolated. The orthogonal plots show very irregular linear decay, but it is nevertheless possible to obtain rough estimates of magnetizations removed by vector subtraction, usually in the interval 15–80 mT or 300–550°C. This removed magnetization is referred to as *B* magnetization. *B* magnetization directions are poorly grouped (triangles of Fig. 4*a*). At temperatures above 600°C the magnetizations commonly show irregular changes in direction and even increase in intensity, probably owing to production of new iron minerals during the heating process. Specimens are noticeably

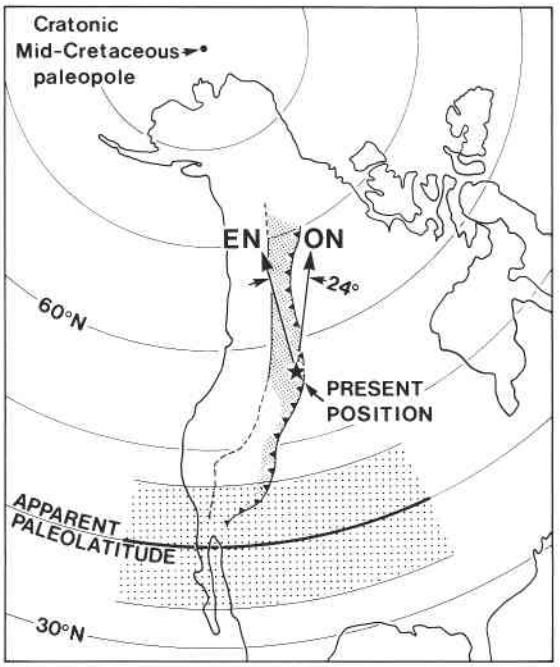


FIG. 6. Paleolatitude and rotation of the Crowsnest Formation *assuming paleosecular variation to have been fully averaged* (see text). The error zones include errors in determining the base map. EN: expected direction of paleonorth at the observing locality if there had been no rotation relative to the craton. ON: observed paleonorth.

redder after heating. Polished sections showed lesser amounts of iron minerals than in *A* specimens. There are irregular particles of magnetite about 10 μm in diameter, and pyrite and chalcopyrite are common. Since time units cannot be recognized, unit weight has been given to each oriented core in the statistical analysis. The cores span the formation stratigraphically. The mean of the *B* magnetizations agrees closely with the 1968 determination (*E* of Table 1; Fig. 4*b*). The mean of the *A* magnetizations is more northerly but not greatly different. *B* and *E* are apparently biassed toward the present Earth's field (PEF of Fig. 4*b*) or its average (the GAD), and we suggest that they contain small, unresolved components of post-Cretaceous age. The ideal behaviour of the *A* magnetizations (Fig. 2) shows that they are good indicators of the paleofield. We therefore regard *A* of Table 1 and Fig. 4 as the best estimate of paleofield recorded by the Crowsnest Formation. However, the following discussion is not substantially affected whether the *A* or *B* magnetization or their combination is used (Table 1).

Interpretation

Paleopoles calculated from the *A* magnetization, both corrected (b) and uncorrected (h) for geological dip, are listed in Table 1 and shown in Fig. 5. They differ from that expected for cratonic North America for the mid-Cretaceous, being to the right and more distant from the sampling locality. Four possible reasons for this anomaly are considered.

The first possibility is that the magnetization is not Albian in age. The direction of *A* corresponds to no known later field, it is very stable (Fig. 2), it shows no evidence of secondary disturbance, and it has the normal polarity appropriate to an Albian age. There is no physical reason to invoke secondary remagnetization. Nevertheless, the possibility requires consideration. The deformation in this part of the fold and thrust belt is Paleocene to Late Eocene, because Paleocene beds of the foothills of the Rockies are folded (Douglas *et al.* 1970, p. 470), whereas Late Eocene to Early Oligocene beds are not (Price 1965). Remagnetization of the Crowsnest Formation may therefore have occurred either in the Late Cretaceous or Paleocene, prior to deformation, or in the Late Eocene or later times, after deformation. The cratonic paleopoles for the former interval of time (70 and 60 Ma, Fig. 5) differ from the Crowsnest Formation paleopole A(b); remagnetization in the Late Cretaceous or Paleocene prior to tilting is improbable. Figure 5 also shows that the paleopole for the Crowsnest Formation without correction for tilt (A(h), Table 1) is far away from the Eocene and later apparent polar wander path for the craton. Likewise, remagnetization in the later Tertiary after tilting is improbable. Hence, there are good reasons for believing the magnetization to be Albian.

TABLE 2. Observed and expected Triassic paleolatitudes of Idaho–Wyoming thrust belt compared

	Paleolatitude (°)
Idaho–Wyoming thrust belt (observed)	
(a) Red beds, possibly Upper Triassic	9 ± 4^a
(b) Red beds, Lower Triassic	8 ± 12^a
(c) (a) + (b)	9 ± 4
Craton (expected at 43.4°N, 110.9°W)	
(d) Wyoming Triassic red beds east of (a) and (b)	8 ± 8^a
(e) Manicouagan impact structure (210 ± 8 Ma), Quebec	$10\pm6^{b,c}$
(f) Chinle Formation, Upper Triassic, New Mexico	10 ± 3^d
(g) State Bridge Formation, Lower Triassic, Colorado	9 ± 3^e
Colorado Plateau (expected at 43.4°N, 110.9°W)	
(h) Moenkopi Formation, L. and M. Triassic, Arizona	13 ± 4^f
(i) Moenkopi Formation, Lower Triassic, Colorado	11 ± 3^g

NOTE: Paleolatitudes expected during the Triassic in the Idaho–Wyoming thrust belt are calculated from data obtained from rock units (d)–(i).

[a] Grubbs and van der Voo (1976).
[b] Robertson (1967).
[c] Larochelle and Currie (1967).
[d] Reeve and Helsley (1972).
[e] Christensen (1974).
[f] Purucker *et al.* (1980).
[g] Helsley and Steiner (1974).

A second possibility is that the rocks are magnetically anisotropic, with a plane of "easy" magnetization in the bedding plane. In order to study this possibility a direct current (DC) field of 1.8 T was applied to four specimens; the field was strong enough to remagnetize them completely. The field was applied approximately parallel to the direction of stable magnetization in each specimen. The orienting device used is accurate to 2° (Irving and Park 1973). The isothermal remanent magnetizations (IRM's) acquired were clustered randomly about the applied field direction, with deviations varying from 1 to 5°. During AF demagnetization up to 100 mT, the directions remained constant, with random deviations from the applied field averaging 3°, much less than the observed mean flattening of 12° (Table 1). It seems, therefore that anisotropy is unlikely to have been responsible for the anomalous direction.

A third possibility is that the effect is tectonic. *If our results span a sufficient time to average out the paleosecular varia-*

tion, the Crowsnest paleopole (A(b)) suggests large-scale motions. The righthandedness of the paleopole indicates that the Crowsnest Formation and the thrust sheets in which it occurs have rotated 24 ± 10° clockwise relative to the craton (Fig. 6). It is not unreasonable to expect that such rotations have occurred. The farsidedness of the paleopole means the inclination is lower than expected and the apparent paleolatitude is 40 ± 6°N, or 17 ± 6° (1800 km) farther south relative to the craton than at present. This effect cannot be caused by any error in correcting for tilt of the beds because the direction of magnetization is essentially perpendicular to the strike (which is very well constrained, Fig. 1), so uncertainties in geological dip could affect the apparent rotation but not apparent paleolatitude. The implication is that the Rocky Mountains of southern Alberta were first carried northwards by almost 2000 km during the interval Albian to Paleocene and then were thrust from the west onto the North American craton. As already noted, this is contrary to the generally accepted geological interpretation, which derives the Rockies from no great distance away along the continental margin (Norris 1964; Price 1981).

The fourth possibility is that the anomaly is caused by insufficient time averaging. The mean of Table 1 may have been deflected from the true average mid-Cretaceous field because substantial nondipole components of the paleofield have not been averaged out. In an attempt to extend the stratigraphic and hence the time coverage we sampled buff-coloured sandstone of the Blairmore Group at the locality noted in Fig. 1 and subjected it to detailed AF and thermal demagnetization. The directions of magnetization were initially random and remained so during demagnetization. No coherent magnetizations are present, and the time coverage remains uncertain. Unless there is ample evidence the paleosecular variation has been averaged out (as is the case, for example, of well-grouped data acquired farther west in the Cordillera from several slowly cooled rock bodies (Irving *et al.* 1985)), there are no firm grounds for invoking large latitudinal motions. We therefore conclude that so far as evidence from the Crowsnest Formation goes, the fourth possibility, although not necessarily correct, is the most reasonable explanation.

Other results from the fold and thrust belt

The Rocky Mountain fold and thrust belt extends southwards into the United States, and if the anomaly we have observed is tectonic in origin, then similar paleomagnetic anomalies should be present there. Results have been obtained from numerous Mesozoic sedimentary units in the Idaho–Wyoming area, from Triassic units in the "upper" younger Prospect thrust sheet, and from Jurassic and Cretaceous units in the older more easterly Darby and Absaroka thrust sheets. The Triassic rocks studied are red beds that have very stable magnetizations, with unblocking temperatures confined to a narrow interval (550–680°C) and with normal and reversed polarities (Grubbs and van der Voo 1976). There is no evidence of secondary overprinting. Results from 12 sites show wide divergences of declination from site to site, indicative of tectonic rotations during emplacement of thrust sheets, but inclinations that are in good agreement. The paleolatitudes are also in excellent agreement with that observed in Triassic red beds from the craton just east of the thrust belt (compare entries (*c*) and (*d*) of Table 2). Also listed in Table 2 are the paleolatitudes expected in the Idaho–Wyoming fold and thrust belt from observations obtained in the craton of North America generally. Agreement is excellent, indicating that the Prospect thrust sheet, although rotated, has not been transported latitudinally by a large amount relative to the craton. There is no flattening of inclination such as we have observed in the Crowsnest rocks.

Results from Jurassic and Cretaceous rocks, in the older, more easterly Darby and Absaroka thrust sheets, have been obtained from siltstones, shales, calcareous shales, and limestones, whose magnetizations are much less stable than those of the Triassic red beds just described (McCabe *et al.* 1982; Schwartz and van der Voo 1984). The results are compared with the cratonic polar path in Fig. 7. Two versions of the cratonic path are given (II and GCO); they are in excellent agreement with one another, even though they have been compiled by different procedures. Rock units studied are the Middle Jurassic Twin Creek Formation (TC), the Upper Jurassic Stump Formation (ST), and four rock units (Smiths and Wayan formations, the Peterson limestone, and the Ephraim conglomerate) ranging from the uppermost Jurassic to the Albian (EC). All have normal polarity. The paleopoles fall on the cratonic path in their expected sequence within the rather large statistical errors, indicating that there has been neither very large translation nor rotation of these thrust sheets relative to the craton. The rotations, evident in the Prospect thrust sheet, are not apparent in the older overlying Darby and Absaroka thrust sheets. The data are unusual because they contain no reversed magnetizations, even though reversals of the paleofield were frequent during the interval of time they were laid down. It is possible, therefore, that the magnetizations were acquired during the Cretaceous Normal Superchron (112–85 Ma), when the field was consistently of normal polarity. As McCabe *et al.* (1982) have demonstrated, restoration of bedding to paleohorizontal in units TC and ST greatly improves their precision, showing that the magnetization predates the Laramide tilting, but this does not preclude a mid-Cretaceous age for the magnetization. Moreover, the magnetization of EC is synchronous with folding, at least in part, because maximum precision (k = 37) is achieved after 60% correction for tilting, whereas full correction yields a lower precision (k = 5). Thus the Early Cretaceous EC rocks were acquiring their magnetization sometime during the Late Cretaceous or earliest Tertiary. Moreover, thermal demagnetization (Fig. 5 of Schwarz and van der Voo (1984); Fig. 2 of McCabe *et al.* (1982)) reveals a wide range of unblocking temperatures (150–650°C) typical of thermal overprints. Such overprinting has recently been well described in not dissimilar rock types of the Brooks Range, Alaska, which apparently possess a remanence acquired during the mid-Cretaceous that replaced earlier remanences (Hillhouse and Grommé 1983). It seems, therefore, that the magnetizations observed in Jurassic and Cretaceous rocks in the Darby and Absaroka thrust sheets cannot be uniquely interpreted. It is possible that the magnetizations of TC and ST are original and the absence of reversals is fortuitous and an artifact of sampling; no motion relative to the craton would then be required. It is also possible that the magnetizations of TC and ST, like that of EC, can be assumed to be secondary and mid-Cretaceous. If this is so, then the TC result indicates a northward translation of 1000 km (Fig. 7). ST and EC show no significant departure from the expected mid-Cretaceous field and hence no motion, but errors are such that substantial motions are not precluded.

Following early work of Collinson and Runcorn (1960) and Norris and Black (1962), numerous results are now available from the Precambrian Belt–Purcell Supergroup of the Rocky Mountains of Alberta and Montana for comparison with the

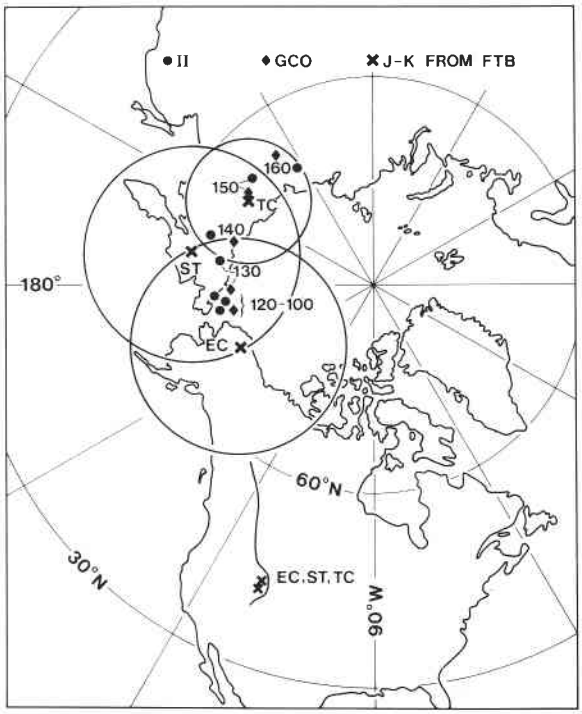


FIG. 7. Paleopoles from Jurassic and Early Cretaceous rocks of the Darby and Absaroka thrust sheets of the fold and thrust belt of the Idaho–Wyoming border region compared with cratonic reference paleopoles. The latter are numbered in millions of years; II is from Irving and Irving (1982), and GCO is from Gordon *et al.* (1984). EC: Early Cretaceous and latest Jurassic (four formations; 69°N, 204°E, A_{95} = 8.7). ST: Stump Formation, Upper Jurassic (64°N, 170°E, A_{95} = 15; Schwarz and van der Voo 1984). TC: Twin Creek Formation, Middle Jurassic (68.4°N, 145°E, A_{95} = 8.7; McCabe *et al.* 1982).

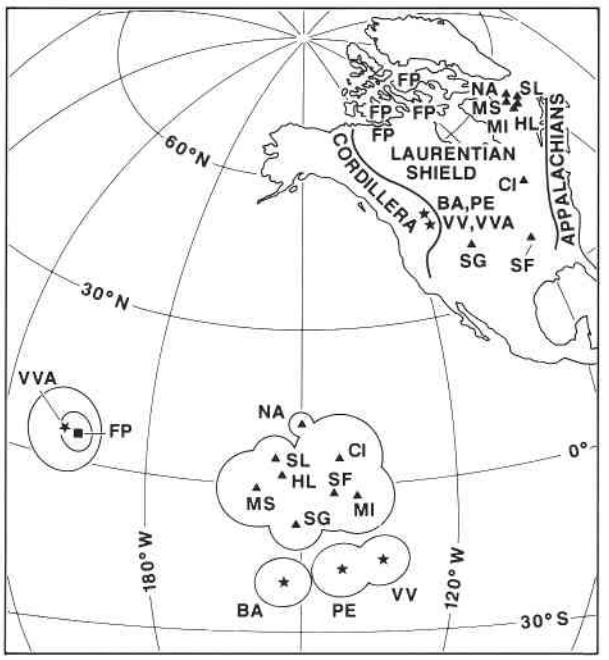


FIG. 8. Mid-Proterozoic (Elsonian) paleopoles from the Laurentian Shield compared with these from the Belt–Purcell Supergroup. PE: average through a substantial thickness of the Belt–Purcell Supergroup in Alberta and Montana obtained by Evans *et al.* (1975). BA: average of 10 formations from the Belt–Purcell of Idaho and Montana (Elston and Bressler 1979). VV: from a study of Spokane Formation by Vitorelli and van der Voo (1977); VVA: is an overprint from the same unit. Data from the Laurentian Shield are as follows. CI: Croker Island Complex (Palmer 1969), 1475 Ma, Rb/Sr isochron; HL: Harp Lake anorthosite (Irving *et al.* 1977), 1450 Ma, U–Pb on zircon; MI: Michikamau anorthosite (Emslie *et al.* 1976), 1460 Ma, U–Pb on zircon; MS: Mistastin pluton (Fahrig and Jones 1976), 1346 Ma, Rb/Sr whole-rock isochron; NA: Nain anorthosite (Murthy 1978), 1418 Ma, Rb/Sr mineral age; SF: St. Francois rocks (Hayes and Scharon 1966), 1400 Ma, Rb/Sr; SG: Sherman granite (Eggler and Larson 1968), 1410 Ma, Rb/Sr isochron; SL: Seal Lake Group (Roy and Fahrig 1973) 1350 Ma, Rb/Sr isochron on associated volcanics. FP: from the Franklin intrusions of northern Canada compiled from Fahrig *et al.* (1971), Robertson and Baragar (1972), Fahrig and Schwarz (1973), and Park (1974).

craton. Six paleopoles from the craton for the interval 1350–1475 Ma form a reasonably good group in the central Pacific triangles of Fig. 8. The age of Belt–Purcell rocks is not accurately fixed, but present evidence indicates that it falls within this range (Obradovich *et al.* 1984). The results of Evans *et al.* (1975) from Alberta (PE, Fig. 8) and Elston and Bressler (1979) from Montana (BA, Fig. 8) were obtained from several formations within the Belt–Purcell. A detailed multicomponent study of the Spokane Formation of Montana by Vitorelli and van der Voo (1977) yields both a paleopole (VV) in good agreement with PE and an overprint (VVA) that agrees with a well-established paleopole from the Franklin intrusives of Late Precambrian age (650–700 Ma). Vitorelli and van der Voo (1977) argued that VVA may relate to a Late Precambrian metamorphic event that is recorded radiometrically in Belt–Purcell rocks. There is, however, considerable uncertainty in the chronology of Late Precambrian paleopoles, and the agreement could be fortuitous. Paleopole BA and, to a lesser extent, PE are spread out toward VVA from VV, and it is possible that they too have been affected to some degree by this later Precambrian event.

Belt–Purcell paleopoles have a small southerly displacement relative to those falling within the same age range from the craton. The displacement could reflect apparent polar wander, but this is unlikely because the paleopole seems to have been remarkably stable during this interval. Alternatively, the difference could be attributed to relative motion since the mid-Proterozoic between the fold and thrust belt and the craton. If this is true, then movement of the Rockies from the north is implied, in the opposite sense to that of Fig. 6. Moreover, if the correlation of the Spokane Formation overprint (VVA) with the Franklinian episode is correct, then the motion must have been accomplished in the Precambrian. However, the differences, although seemingly significant, are not large and indicate that the basement to the present-day Rockies has been in the vicinity of western North America since the mid-Proterozoic. It is also possible that the Belt–Purcell rocks studied are younger than 1350 Ma, and the above comparisons may have no tectonic relevance.

Only results from the Twin Creek Formation (assuming, for reasons given above, that its magnetization is mid-Cretaceous) can be interpreted as indicative of northward Laramide displacement of the fold and thrust belt, and this interpretation is insecure. All other paleomagnetic data, especially those from Triassic rocks (Table 2), indicates no large northerly motion.

We therefore conclude that the balance of the present evidence favours the hypothesis that the anomalous paleolatitude record of the Crowsnest Formation arises because it contains only a short record of the paleofield. The entire unit, 320 m

thick, could have been laid down and magnetized in tens rather than the thousands of years required to average the paleosecular variation. The fact that the *apparent* northward displacement of the Crowsnest Formation is roughly similar in form to that observed in other Cretaceous rock units farther west in the Cordillera (Irving *et al.* 1985) has, on this view, to be regarded as fortuitous. The apparent rotation (24 ± 10°) also could be an artifact of an incomplete sampling of the paleosecular variation, but it could reflect a real tectonic rotation.

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